MEF-RANK: MODEL-AWARE ACTION MEMORY FOR FINANCIAL RANKING

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ABSTRACT

Historical feedback can improve financial ranking when its corrections match both market conditions and the predictor's decision state. We propose MEF-Rank, a model-aware action memory for selectively reusing such feedback across heterogeneous predictors. An offline Writer distills matured outcomes into screened adjustments of seven financial signal groups. Joint market–model retrieval identifies compatible experience, while uncertainty calibration and a bounded inverse map translate its action into the current predictor's gate coordinates. The predictor recomputes its representations with frozen weights, or returns its original prediction exactly when support is inadequate. Across ten adapted backbones and five markets, MEF-Rank improves mean RankIC in all 50 pairs, raising the macro average from 0.052161 to 0.063573, a 21.9% relative gain. Targeted controls support the value of model-state conditioning and structured action retrieval, demonstrating the framework's broad applicability to financial ranking without test-time parameter updates.

***Index Terms*—** Financial ranking, model-aware retrieval, action memory, historical feedback, time-series forecasting

1. INTRODUCTION

Financial ranking uses price, liquidity, and risk signals whose relevance changes with market conditions. Historical outcomes can reveal useful corrections, but their effects depend on the predictor's current signal use: the same gate offset behaves differently near saturation. Reusing historical feedback therefore requires both selecting compatible experience and realizing its intended utilization change.

Existing work addresses non-stationarity through instance normalization [1], financial data and model adaptation [2], and gradient-conditioned layer transformations with associative memory (FSNet) [3]. Retrieval supplies examples for language modeling [4] and future patches for forecasting (RAFT) [5]. ResMem corrects outputs with retrieved residuals [6]; the recent RATL preprint also uses frozen-model representations to retrieve model-specific residuals [7]. MEF-Rank formulates experience as *mean changes in signal utilization*, constructed with nested held-out screening and realized through inversion of the current gates.

A first pass exposes signal utilization, representation disagreement, and output uncertainty as model state B, alongside market state J. Joint retrieval selects compatible experience; calibrated support determines how much to reuse. The retrieved action specifies seven groupwise mean gate changes, which a bounded inverse converts into current logit offsets. Unsupported queries retain the cached first-pass output exactly.

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We introduce a model-aware action memory coupling screened utilization shifts, joint-state retrieval, and current-gate inversion. A shared seven-group interface enables this mechanism across heterogeneous backbones, each with its own memory. All 50 backbone–market pairs improve in mean RankIC, yielding a 21.9% macro-average gain and positive paired intervals in all five markets. Controls support model-state conditioning and structured retrieval beyond static task adaptation. Code and configurations: `https://github.com/IFIFIF23/MEF-Rank`.

2. MODEL-AWARE EXPERIENCE FEEDBACK

2.1. A shared interface for heterogeneous predictors

At decision cutoff c_t, masked inputs $X_t \in \mathbb{R}^{N_t \times 60 \times 125}$ contain available observations for N_t assets. Seven groups cover trend, liquidity, downside risk, valuation/size, reversal/style, market-relative signals, and price paths (six groups of 8 features and one of 77). The H5 ranking target is the adjusted-open return from session $t+1$ to $t+6$; equity decisions follow the close and crypto uses UTC daily boundaries. Labels become available only after their exit timestamp.

For group representation h_{tig} and gate logit z_{tig}, a shared date-level action $\beta \in \mathbb{R}^7$ changes signal utilization:

$$r_{tig}(\beta_g) = \sigma(z_{tig} + \beta_g), \qquad \widetilde{h}_{tig} = r_{tig}(\beta_g) h_{tig}. \tag{1}$$

Hooks act on group-aligned sequence, variable, or patch representations, after normalization that could cancel scaling and before affected backbone operations; missing groups remain masked. A shared task head combines a 20-bin ordinal expectation and bounded residual, $s_{ti} = \sum_q p_{tiq} v_q + 0.1 \tanh(e_{ti})$, with fixed bin centers v_q and ordinal/pairwise ranking losses. Training uses $\beta = 0$. The first-pass prediction is $s_t^0 = F_\theta(X_t; 0)$ (Off), retaining the trained static gates and task head. All subsequent feedback keeps θ fixed.

2.2. Model state: retrieving for the current decision

The first pass supplies the 37-dimensional diagnostic state

$$B_t = [R_{\text{mean}}(7), R_{\text{std}}(7), E(7), D(7), U(5), P(4)]. \tag{2}$$

R_{mean} and R_{std} summarize each group's zero-action gate values over valid assets. E gives representation-exposure proxies; D captures between-group representation disagreement on shared valid assets. U summarizes output entropy, score dispersion, top-score gaps, and near ties; P summarizes matured prequential performance.

The market descriptor $J_t \in \mathbb{R}^{306}$ summarizes levels, velocities, and curvatures over 5/20/60 days. Both descriptors use availability masks; label-dependent coordinates obey the frozen information cutoff, and unavailable coordinates, including unsupported P, remain masked.

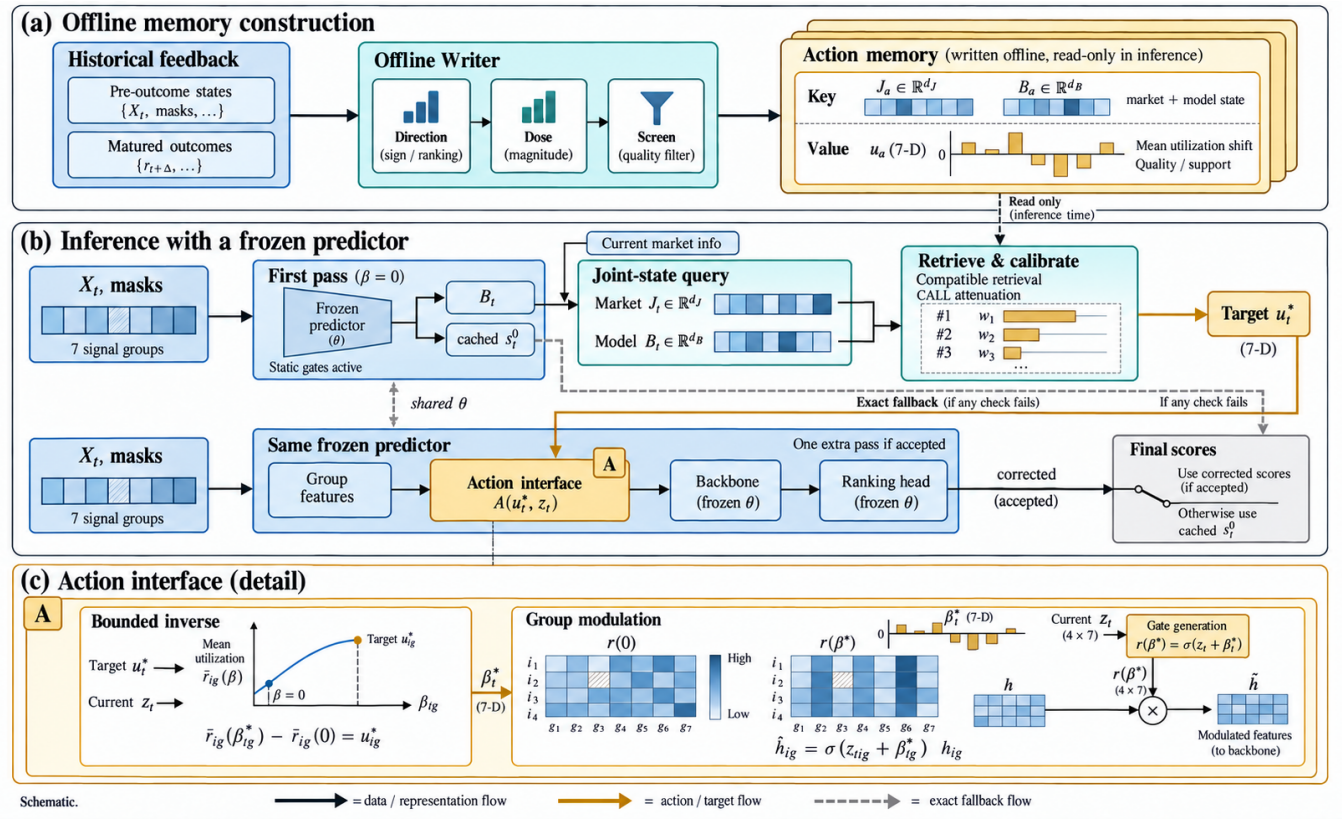


Fig. 1. MEF-Rank. (a) Historical feedback is distilled into action memory. (b) Joint market–model retrieval and offline-fitted calibration select a utilization target for the frozen predictor. (c) Bounded inversion and group modulation produce one corrected pass; failed checks return the cached output.

2.3. Distilling historical outcomes into reusable actions

In Fig. 1(a), historical checkpoints and preprocessing are fitted before each quarterly evaluation Episode. An anchor's first-pass state is recorded out of sample; the Writer uses its outcome only after label maturity, with the predictor frozen. Twenty asset splits, stratified by size and liquidity, allocate 49%/21%/30% to direction search, dose selection, and held-out screening. Only supervision is split: every forward pass retains the full cross-sectional feature context. Direction search jointly optimizes β using five projected-gradient steps on a K-weighted ordinal/pairwise surrogate (initial step 0.15, logit cap 0.75, regularizer $0.05\|\beta\|_2^2$).

With universe size N, decision depth $K = 50$, and subset S, $|S| > 1$, let $k_S = \min\{|S|, \max(1, \lfloor |S| \min(K, N)/N + 0.5 \rfloor)\}$ and $\alpha_S = (|S| - k_S)/(|S| - 1)$. Dose and screening use

$$U_S = \alpha_S \,\mathrm{NDCG@}k_S + (1 - \alpha_S)(1 + \mathrm{RankIC}_S)/2, \quad (3)$$

where NDCG [8] uses relevance $2^{y_i^S} - 1$ and y_i^S is the subset-local return percentile rank. For $N \le K$, the utility reduces to the RankIC endpoint. Each nonzero utilization shift from Eq. (4) defines a unit direction; zero shifts admit only the zero candidate. Dose selection chooses the smallest feasible shift norm reaching 95% of the best positive utility gain over zero, or zero if no candidate improves it. Feasible doses must be invertible and have nonnegative dose-subset gains in RankIC, fractional Precision@k, and mean top-k return. Acceptance requires at least 16 technically valid repeats, positive median outer utility gain, and nonnegative median outer gains in RankIC, fractional Precision@k, and mean top-k return. All valid repeat actions, including zeros and negative-gain cases, enter screening. Their mean defines a unit consensus when nonzero; their median nonnegative projection onto it fixes the dose. Zero consensus or dose rejects the record. Full-label refitting changes only the direction, requiring cosine at least 0.5 to that consensus; feasibility can only reduce the dose. Screening evaluates the repeat actions, not the refitted action on an independent sample.

The memory stores the resulting mean *utilization shift*

$$u_{ag} = \frac{1}{|V_{ag}|} \sum_{i \in V_{ag}} [\sigma(z_{aig} + \beta_{ag}) - \sigma(z_{aig})], \quad (4)$$

where V_{ag} contains valid assets; $u_{ag} = 0$ for empty V_{ag}. Each record pairs (J_a, B_a) with u_a, quality, support, and temporal and identity metadata. Its 7×7 covariance C_a is the Ledoit–Wolf shrinkage covariance of all valid repeat utilization shifts, repaired to be positive semidefinite and frozen before full-label refitting. Availability requires label maturity and Writer completion on the historical replay timeline.

2.4. Selective retrieval and uncertainty calibration

Historical median/MAD scaling and blockwise Ledoit–Wolf Mahalanobis distances use common valid coordinates (minimum fraction 0.50), with $d(t, a) = 0.60 d_J(t, a) + 0.40 d_B(t, a)$. Time, market, backbone, seed, label and action compatibility, support, and out-of-distribution checks filter candidates. The nearest record per quarterly

Table 1. Same-checkpoint Off/On RankIC: mean ± sample SD over three training seeds, with all fallback days included. Values are unscaled; bold denotes the higher mean in each Off/On pair.

Backbone	Mode	gem	main	Nasdaq-100	S&P 500	Crypto-15
STR	Off	.053422±.000951	.057143±.000966	.056727±.001096	.054629±.001489	.044794±.002835
	On	**.062735**±.000737	**.067009**±.001475	**.063474**±.000765	**.065086**±.001317	**.057921**±.001508
iTransformer	Off	.052919±.001090	.042716±.000917	.054782±.001718	.050847±.000801	.048957±.002986
	On	**.060257**±.000788	**.062556**±.001212	**.062283**±.001139	**.064776**±.001500	**.054684**±.002774
TimeMixer	Off	.057185±.000729	.061537±.000937	.060418±.001226	.058234±.000621	.041846±.002750
	On	**.065515**±.001316	**.069993**±.001274	**.067700**±.001302	**.067026**±.001370	**.047391**±.001923
ModernTCN	Off	.061946±.001159	.054386±.001147	.064238±.001409	.062317±.001036	.055392±.001993
	On	**.069417**±.000790	**.067191**±.000659	**.073774**±.001397	**.073092**±.000725	**.058816**±.001649
SOFTS	Off	.053566±.001265	.058624±.000999	.056327±.001529	.054109±.000859	.045926±.002009
	On	**.068911**±.001328	**.067544**±.001001	**.069175**±.000794	**.069007**±.001288	**.057631**±.002896
TimeXer	Off	.044916±.000829	.046951±.001038	.047593±.000975	.048735±.000624	.050613±.003041
	On	**.057291**±.001408	**.060974**±.001285	**.060164**±.001567	**.061006**±.001189	**.057284**±.002265
TimeBridge	Off	.051066±.001393	.063175±.001186	.053814±.001629	.044271±.000503	.049874±.002493
	On	**.064662**±.000595	**.072222**±.000805	**.068790**±.001147	**.055227**±.000578	**.056725**±.002535
TimeFilter	Off	.043117±.000563	.052693±.000771	.044452±.001604	.050392±.000737	.035748±.002094
	On	**.062257**±.000999	**.068049**±.000944	**.067483**±.001035	**.066744**±.000599	**.046932**±.001310
SRSNet	Off	.051976±.000528	.056841±.000673	.055619±.000843	.052481±.000737	.049735±.002301
	On	**.065865**±.000514	**.067798**±.000955	**.064766**±.001411	**.066245**±.001058	**.055842**±.002751
ChannelTokenFormer	Off	.049181±.000766	.049527±.000609	.051842±.001074	.052841±.000633	.047628±.001672
	On	**.065834**±.000757	**.066841**±.000852	**.066207**±.000886	**.066713**±.001151	**.053741**±.001509

Episode competes for the top five places, with at least two effective Episodes.

Normalized weights combine $\exp(-d(t,a)/\tau)$, support, quality, and covariance-based precision. For $\mu_t = \sum_a w_a u_a$, the target is

$$u_t^* = \gamma_t \mu_t, \qquad \Sigma_t = \sum_a w_a^2 [(u_a - \mu_t)(u_a - \mu_t)^\top + C_a]. \quad (5)$$

CAL1 uses leave-one-Episode-out historical calibration with at least 20 reference Episodes. Its attenuation combines coverage, quality, and confidence:

$$\gamma_t = \kappa_t \eta_t \xi_t, \quad \kappa_t = [M_t / M_{\text{ref}}]_0^1,$$
$$\eta_t = \left[\frac{\sum_a w_a \min(g_a, g_{\text{cap}})}{g_{\text{ref}} + \epsilon} \right]_0^1, \quad \xi_t = \frac{\|\mu_t\|_2^2}{\|\mu_t\|_2^2 + \operatorname{tr} \Sigma_t + \epsilon}. \quad (6)$$

Here $[x]_0^1 = \min(1, \max(0, x))$, M_t is kernel mass, g_a is median outer utility gain, and $\epsilon = 10^{-12}$. Episode-balanced calibration sets the reference quantities and gain cap. Since $\gamma_t \in [0,1]$, uncertain or weakly supported evidence is attenuated. Effective support $1/\sum_a w_a^2 \geq 2$ and disagreement gates further restrict execution; Σ_t is an operational uncertainty score.

2.5. Transporting the action to the current predictor

A historical logit offset can produce a different effect when current gates saturate differently. MEF-Rank instead preserves the intended utilization change. For nonempty V_{tg}, let $\bar{r}_{tg}(b) = |V_{tg}|^{-1} \sum_{i \in V_{tg}} \sigma(z_{tig} + b)$. The interface in Fig. 1(c) solves

$$\bar{r}_{tg}(\beta_{tg}^*) - \bar{r}_{tg}(0) = u_{tg}^*, \qquad \beta_{tg}^* \in [-0.5, 0.5]. \quad (7)$$

For finite logits, $\bar{r}'_{tg}(b) > 0$, so a unique solution exists exactly when u_{tg}^* lies between $\bar{r}_{tg}(-0.5) - \bar{r}_{tg}(0)$ and $\bar{r}_{tg}(0.5) - \bar{r}_{tg}(0)$. Unsupported groups require zero targets and are skipped; a nonzero target causes fallback. Bisection uses 24 float64 steps and residual tolerance 10^{-5}. All coordinates must pass feasibility checks; unreachable targets are rejected. Accepted actions trigger one additional pass through affected backbone operations, yielding $F_\theta(X_t; \beta_t^*)$ (On). Any failed check returns cached s_t^0 exactly.

3. EXPERIMENTS

3.1. Data, backbones, and paired evaluation

We evaluate ChiNext (gem), China's main board (main), Nasdaq-100, S&P 500, and Crypto-15, using Tushare for Chinese equities and Yahoo Finance for US equities and crypto. Historical membership snapshots define daily equity universes; crypto uses 15 fixed assets. The five markets contain 601/601/624/624/914 valid test dates and median daily universe sizes of 1332/3056/97/472/15. Missing observations are masked and corporate-action adjustments follow the data providers.

Predictors train on 2018–2022. Memory, CAL1, thresholds, and label-dependent states use only information available by end-2022. The 2023 period checks the fixed MEF system; Residual-kNN separately uses it to select correction strength. Test data start in 2024, with raw prices ending on 8 July 2026. The last prediction dates with complete H5 labels are 30 June (China), 29 June (US), and 2 July 2026 (crypto). Boundary-crossing labels are purged and fitted preprocessing is frozen. Historical experience anchors span 2015–2022 for gem/crypto and 2013–2022 for main/US. Initial gem/crypto historical checkpoints train on 2010–2014 data, preceding their 2015 anchors.

The ten backbones comprise a Spatio-Temporal Ranker (STR), with a temporal–cross-sectional Transformer motif related to MASTER [9], and iTransformer [10], TimeMixer [11], ModernTCN [12], SOFTS [13], TimeXer [14], TimeBridge [15], TimeFilter [16], SRSNet [17], and ChannelTokenFormer [18].

Table 2. Representative controls: RankIC mean ± sample SD over three training seeds. I: iTransformer; M: ModernTCN; G: gem; S: S&P 500. Task-only is independently trained; Off/On repeat Table 1.

Setting	I–G	I–S	M–G	M–S
Task-only	.053092±.001061	.051440±.001140	.062917±.000582	.063194±.000901
MEF-Off	.052919±.001090	.050847±.000801	.061946±.001159	.062317±.001036
MEF-On (Full)	**.060257**±.000788	**.064776**±.001500	**.069417**±.000790	**.073092**±.000725
J-only	.053984±.000963	.052381±.000799	.063013±.001278	.063249±.001263
Random retrieval	.052783±.000872	.054107±.001242	.062077±.001176	.062902±.001086
Residual-kNN	.053497±.001406	.053986±.000954	.061868±.000574	.063682±.001012

Table 3. Market-level paired gains and 20-day-block bootstrap 95% intervals. APPLY is the fraction of valid queries receiving a correction, pooled across ten backbones and three seeds.

Market	ΔRankIC	95% interval	APPLY
gem	.012345	[.011020,.013894]	83.72%
main	.012658	[.011267,.014160]	86.36%
Nasdaq-100	.011800	[.010513,.013227]	73.16%
S&P 500	.012607	[.011323,.013862]	83.65%
Crypto-15	.007645	[.004215,.011254]	34.23%

Each backbone–market–seed has its own predictor and memory. Off/On share weights, static interfaces, preprocessing, assets, and dates, isolating the effect of enabling historical feedback on an adapted predictor.

We report daily cross-sectional Spearman RankIC, averaged over all valid dates including fallback dates. Means and sample SDs use training seeds 20260921/20260922/20260923. Ties receive average ranks; constant predictions/labels and insufficient-asset dates are invalid, with zero such exclusions in the reported evaluation.

3.2. Consistent gains across architectures and markets

Table 1 shows positive mean gains in all 50 backbone–market pairs. The equal-pair macro RankIC increases from 0.052161 to 0.063573: an absolute gain of 0.011411 and a 21.9% relative increase in the macro average. The 40 equity pairs average a 0.012353 gain, while crypto averages 0.007645. Gains span attention, mixing, convolutional, and graph-based backbones, demonstrating the shared interface's effectiveness across diverse computational structures.

All five market-level 95% intervals in Table 3 exclude zero. They use daily paired On–Off differences averaged over seeds and then backbones, with 2000 moving-block bootstrap resamples [19], 20-day blocks, and seed 31001. Identical sampled date blocks preserve cross-backbone dependence; 10-/40-day sensitivity intervals also remain positive. These results support consistent market-level gains under temporal block resampling, conditional on the trained seeds.

3.3. Benefits of model-state conditioning and retrieval

Table 2 tests iTransformer and ModernTCN on gem and S&P 500. *J-only* removes B from distance and B-dependent screening/calibration, recalibrating on the same history. Full MEF-Rank exceeds it by 0.006273–0.012395 across the four settings, supporting model-state conditioning in experience selection and calibration.

Random selects from the same eligible Episode pool, retaining weighting, calibration, and fallback. Retrieval seeds 4101–4105 are averaged at the metric level within each training seed. *Residual-kNN*, inspired by ResMem [6], retrieves out-of-sample percentile-rank residuals using standardized input-window keys and the same history range, without Writer acceptance filtering. Its correction strength is selected on 2023 validation. Full exceeds Random by 0.007340–0.010669 and Residual-kNN by 0.006760–0.010790. The residual comparison assesses complete systems, with different keys, memory filtering, and correction spaces.

Task-only retains financial inputs, masks, head, and losses but omits MEF gates, and is independently trained under matched rules. Full exceeds Task-only by 0.006500–0.013336, while Off is 0.000173–0.000971 lower. Feedback thus delivers gains beyond the static interface. Across all four settings, Full has the highest mean RankIC among all compared variants. These control comparisons concern mean differences over seeds.

3.4. Selective execution and computational cost

Execution rates (Table 3) range from 73.16% to 86.36% on equities and reach 34.23% on crypto; all remaining queries preserve Off exactly. Gains are measured over all dates, so the reported benefit already accounts for this selective use of history. Eligible calibration support ranges from 22 to 40 Episodes across runs, above the required 20.

On gem with seed 20260921, iTransformer and ModernTCN each use 1,500 memory records (about 4 MB, excluding predictor weights). Complete daily Off/On latency is 146/197 ms and 334/461 ms (34.9%/38.0% overhead) on an A800 80-GB GPU (FP32, PyTorch 2.6.0+cu124). Timing includes data loading and all feedback stages, using 100 evenly spaced distinct test dates measured five times, 20 additional warm-ups, and GPU synchronization. Task-only takes 90/290 ms; Residual-kNN takes 207/421 ms with 210/463 MB memories. Historical checkpoint training, Writer, and calibration require about 41.5/53.0 h of additional preparation per configuration, versus 0.89/1.12 h for shared predictor training. These are wall-clock times.

4. CONCLUSION

MEF-Rank turns historical feedback into model-compatible internal actions through joint-state retrieval, utilization-space memory, and current-gate inversion. Its shared interface improves mean RankIC in all 50 backbone–market pairs, with a 21.9% macro-average gain and positive paired intervals in all five markets. Targeted controls support model-state conditioning and structured historical reuse. Together, these results demonstrate the effectiveness of the complete framework across diverse adapted predictors without test-time parameter updates.

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